Gauss Elimination, RREF, and Inverses

Linear Algebra · Lecture 2

Last time (Lecture 1): products of matrices and their rules; the columns and rows of a product; the transpose. e_i is column i of the identity matrix (I_m or I_n, as the product requires).

1. Linear systems and row operations

Definition 1. Let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. The system of m linear equations $a_{i1}x_1 + \cdots + a_{in}x_n = b_i$ $(i = 1, \ldots, m)$ in n unknowns is $Ax = b$ (Lecture 1, Definition 2). A is the **coefficient matrix**, $[A \mid b] \in \mathbb{R}^{m\times(n+1)}$ the **augmented matrix**. A **solution** is an $x \in \mathbb{R}^n$ with $Ax = b$. The system is **consistent** if it has a solution, **inconsistent** otherwise; **homogeneous** if $b = 0$, and then $x = 0$ is a solution, the **trivial solution**.

Definition 2. The **elementary row operations** on a matrix with m rows are, for rows $r \neq s$: (1) $R_r \leftarrow cR_r$ $(c \neq 0)$: multiply row r by c; (2) $R_r \leftarrow R_r + cR_s$ $(c \in \mathbb{R})$: add c times row s to row r; (3) $R_r \leftrightarrow R_s$: interchange rows r and s. Several operations on one arrow are performed in the order written (upper label first). B is **row-equivalent** to A if finitely many elementary row operations take A to B.

An operation computes each column of the new matrix from the same column of the old one: operations that take $[A \mid b]$ to $[B \mid z]$ take A to B, and operations that take A to R take $[A \mid 0]$ to $[R \mid 0]$.

Theorem 3. (a) Each elementary row operation is undone by one of the same type: $R_r \leftarrow cR_r$ by $R_r \leftarrow c^{-1}R_r$, $R_r \leftarrow R_r + cR_s$ by $R_r \leftarrow R_r - cR_s$, $R_r \leftrightarrow R_s$ by itself.

(b) If row operations take $[A \mid b]$ to $[B \mid z]$, then $Ax = b$ and $Bx = z$ have the same solutions.

Proof.

(a) Row by row; the other rows do not change:

$$c^{-1}(c\,\mathrm{row}_r) = \mathrm{row}_r$$
$$(\mathrm{row}_r + c\,\mathrm{row}_s) - c\,\mathrm{row}_s = \mathrm{row}_r \qquad \text{(row } s \text{ is unchanged)}$$
two interchanges of rows r, s restore their order

(b) It suffices to take one operation:

$$Ax = b \implies Bx = z \qquad \text{(each equation of } Bx = z \text{ is one of } Ax = b, c \text{ times one, or one plus } c \text{ times another)}$$
$$Bx = z \implies Ax = b \qquad \text{((a): an operation takes } [B \mid z] \text{ back to } [A \mid b])$$

□

2. Reduced row echelon form

Definition 4. A matrix R is in **reduced row echelon form** (**RREF**) if

(a) the first nonzero entry of each nonzero row is 1, its **pivot**;

(b) each column that contains a pivot has all its other entries 0;

(c) every zero row lies below every nonzero row;

(d) if rows $1, \ldots, r$ are the nonzero rows and the pivot of row i is in column k_i, then $k_1 < \cdots < k_r$.

Columns $k_1, \dots, k_r$ are the **pivot columns**; column k_i of R is e_i (by (a), (b)). A matrix that satisfies (a), (c), (d) is in **echelon form**; then every entry below a pivot is 0 (by (c), (d)).

For example, I_n and 0 are in RREF, and so is $R = \left(\begin{smallmatrix} 0 & 1 & -3 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{smallmatrix}\right)$ ($r = 2$, $k_1 = 2$, $k_2 = 4$). $\left(\begin{smallmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{smallmatrix}\right)$ is in echelon form, not in RREF (column 3 violates (b)); $\left(\begin{smallmatrix} 0 & 2 & 1 \\ 1 & 0 & -3 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ violates (a) and (d).

Definition 5 (Gauss–Jordan elimination)**.** Let M be any matrix. Start with $i = 1$.

Step 1. Find the first column with a nonzero entry in rows $i, i+1, \dots$ (the next pivot column); if there is none, go to Step 5.

Step 2. If row i has 0 in this column, interchange it with a lower row that has a nonzero entry there.

Step 3. Multiply row i by the reciprocal of this entry, so that the pivot is 1.

Step 4. Add multiples of row i to the rows below it to make every entry below the pivot 0. Increase i by 1; return to Step 1.

Step 5. (**Backward elimination**.) For each pivot, from the last to the first, add multiples of its row to the rows above it to make every entry above the pivot 0.

Steps 1–4 (**forward elimination**) produce an echelon form, Step 5 a matrix in RREF:

- When Step 1 finds the pivot column of row i, rows $i, i+1, \dots$ are 0 in every column before it, and Step 4 makes rows $i+1, i+2, \dots$ 0 in it: each nonzero row begins with its pivot 1, the pivot columns are taken from left to right, the rows left when Step 1 finds none are zero; an echelon form, pivots in columns $k_1 < \cdots < k_r$.
- In Step 5, when the pivot of row i is processed, row i is 0 before column k_i and in columns $k_{i+1}, \dots, k_r$ (done): the multiples of row i change no entry before column k_i and no other pivot column. In the end every column k_i is e_i.
- Step 3 may be postponed, to the end of forward elimination or to the very end (the pivots become 1 only then); this often avoids fractions.

So every matrix M has an **RREF**: a matrix in RREF that is row-equivalent to M.

Example 6. Solve $x_1 + x_2 + x_3 - 2x_4 = 0$, $x_1 + x_2 + 2x_3 - 3x_4 = 0$, $2x_1 + x_2 + 5x_3 - 4x_4 = 0$.

Solution.

Row-reduce the coefficient matrix A (the column $b = 0$ never changes; not written):

$$\begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & 2 & -3 \\ 2 & 1 & 5 & -4 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & -1 & 3 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 + 3R_3} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

Entry $(2,2)$ is 0 after the first arrow: Step 2 interchanges rows 2, 3; Step 3 multiplies row 2 by -1. The first matrix of the second line is an echelon form.

The last matrix R is in RREF, pivot columns $1, 2, 3$; $Rx = 0$ has the solutions of $Ax = 0$ (Theorem 3(b)):

$$x_1 + 2x_4 = 0, \quad x_2 - 3x_4 = 0, \quad x_3 - x_4 = 0; \qquad x_4 = t \text{ free}: \quad x = t(-2, 3, 1, 1)' \quad (t \in \mathbb{R})$$

Check: $-2 + 3 + 1 - 2 = 0$, $-2 + 3 + 2 - 3 = 0$, $-4 + 3 + 5 - 4 = 0$.

Proposition 7. Let row operations take $[A \mid b]$ ($A \in \mathbb{R}^{m\times n}$) to $[R \mid z]$, R in RREF with nonzero rows $1,\dots,r$ and pivot columns $k_1 < \cdots < k_r$; $x_{k_1},\dots,x_{k_r}$ are the **pivot unknowns**, the other $n-r$ unknowns the **free unknowns**.

(a) $Ax = b$ is consistent $\iff$ $z_i = 0$ for every $i > r$ (no row $(0\ \cdots\ 0 \mid z_i)$, $z_i \neq 0$).

(b) If so, all solutions: the free unknowns arbitrary, $x_{k_i} = z_i - \sum_{j \text{ free}} R_{ij}x_j$ $(1 \leqslant i \leqslant r)$.

Proof. $Ax = b$ and $Rx = z$ have the same solutions (Theorem 3(b)); column k_l of R is e_l (Definition 4).

(a) Equation $i > r$ of $Rx = z$ reads $0 = z_i$:

$z_i \neq 0$ for some $i > r$ $\implies$ no solution

$z_i = 0$ for all $i > r$: take the free unknowns 0 and $x_{k_l} = z_l$ $(l \leqslant r)$

$\implies Rx = \sum_{l=1}^{r} z_l \operatorname{col}_{k_l}(R) = \sum_{l=1}^{r} z_l e_l = z$ (Lecture 1, Theorem 4(b))

(b) x_{k_i} occurs only in equation i, with coefficient 1; equations $i > r$ read $0 = 0$:

equation $i \leqslant r$: $\quad x_{k_i} + \sum_{j \text{ free}} R_{ij}x_j = z_i$

$\implies$ each choice of the free unknowns gives exactly one solution, and every solution arises so

□

Example 8. Let $A = \left(\begin{smallmatrix} 1&2&0&1\\ 2&3&1&1\\ 1&1&1&1\\ 2&2&2&1 \end{smallmatrix}\right)$. For which $y = (y_1,y_2,y_3,y_4)'$ is $Ax = y$ consistent? Solve it for $y = (4,7,4,7)'$ and for $y = (1,0,0,0)'$.

Solution.

Forward elimination of $[A \mid y]$, keeping $y_1,\dots,y_4$ as letters:

$$\left(\begin{array}{cccc|c} 1&2&0&1&y_1\\ 2&3&1&1&y_2\\ 1&1&1&1&y_3\\ 2&2&2&1&y_4 \end{array}\right) \xrightarrow[R_4\leftarrow R_4-2R_1]{R_2\leftarrow R_2-2R_1,\ R_3\leftarrow R_3-R_1} \left(\begin{array}{cccc|c} 1&2&0&1&y_1\\ 0&-1&1&-1&y_2-2y_1\\ 0&-1&1&0&y_3-y_1\\ 0&-2&2&-1&y_4-2y_1 \end{array}\right)$$

$$\xrightarrow{R_2\leftarrow -R_2} \left(\begin{array}{cccc|c} 1&2&0&1&y_1\\ 0&1&-1&1&2y_1-y_2\\ 0&-1&1&0&y_3-y_1\\ 0&-2&2&-1&y_4-2y_1 \end{array}\right) \xrightarrow[R_4\leftarrow R_4+2R_2]{R_3\leftarrow R_3+R_2} \left(\begin{array}{cccc|c} 1&2&0&1&y_1\\ 0&1&-1&1&2y_1-y_2\\ 0&0&0&1&y_1-y_2+y_3\\ 0&0&0&1&2y_1-2y_2+y_4 \end{array}\right)$$

The last step of forward elimination (the left part is now an echelon form); then backward elimination:

$$\xrightarrow{R_4\leftarrow R_4-R_3} \left(\begin{array}{cccc|c} 1&2&0&1&y_1\\ 0&1&-1&1&2y_1-y_2\\ 0&0&0&1&y_1-y_2+y_3\\ 0&0&0&0&y_1-y_2-y_3+y_4 \end{array}\right) \xrightarrow[R_1\leftarrow R_1-R_3]{R_2\leftarrow R_2-R_3} \left(\begin{array}{cccc|c} 1&2&0&0&y_2-y_3\\ 0&1&-1&0&y_1-y_3\\ 0&0&0&1&y_1-y_2+y_3\\ 0&0&0&0&y_1-y_2-y_3+y_4 \end{array}\right)$$

$$\xrightarrow{R_1\leftarrow R_1-2R_2} \left(\begin{array}{cccc|c} 1&0&2&0&-2y_1+y_2+y_3\\ 0&1&-1&0&y_1-y_3\\ 0&0&0&1&y_1-y_2+y_3\\ 0&0&0&0&y_1-y_2-y_3+y_4 \end{array}\right)$$

Read off (Proposition 7): the left part is an RREF of A (Definition 2), $r = 3$, pivot columns $1,2,4$, $z_4 = y_1 - y_2 - y_3 + y_4$:

consistent $\iff$ $y_1 - y_2 - y_3 + y_4 = 0$; then, with $x_3 = c$ free ($c \in \mathbb{R}$):

$x_1 = -2y_1 + y_2 + y_3 - 2c, \quad x_2 = y_1 - y_3 + c, \quad x_3 = c, \quad x_4 = y_1 - y_2 + y_3$

$y = (4,7,4,7)'$: $4 - 7 - 4 + 7 = 0, \quad x = (3,0,0,1)' + c(-2,1,1,0)'$

$y = (1,0,0,0)'$: $1 - 0 - 0 + 0 = 1 \neq 0, \quad$ no solution

Check: $A(3,0,0,1)' = (4,7,4,7)'$; $A(-2,1,1,0)' = 0$, that is, $\operatorname{col}_3(A) = 2\operatorname{col}_1(A) - \operatorname{col}_2(A)$.

Theorem 9. (a) If $A \in \mathbb{R}^{m\times n}$ with $m < n$, then $Ax = 0$ has a nontrivial solution.

(b) Let $A \in \mathbb{R}^{n\times n}$ and R an RREF of A. Then $Ax = 0$ has only the trivial solution $\iff R = I_n$.

Proof. Let R be an RREF of A (Definition 5), with r nonzero rows; the operations take $[A \mid 0]$ to $[R \mid 0]$ (Definition 2).

(a) Proposition 7(b):

$$r \leqslant m < n \implies \text{a free unknown exists; it } = 1, \text{ the others } = 0 \implies \text{a solution } x \neq 0$$

(b) Theorem 3(b), Proposition 7(b); Lecture 1, Theorem 3(a):

$$\begin{aligned} R = I &\implies Rx = x \implies x = 0 \text{ is the only solution} \\ \text{only } x = 0 &\implies \text{no free unknown (one would give } x \neq 0) \implies r = n \\ &\implies 1 \leqslant k_1 < \cdots < k_n \leqslant n \implies k_i = i,\ \mathrm{col}_i(R) = e_i \implies R = I \quad \text{(Definition 4)} \end{aligned}$$

□

3. Inverses and elementary matrices

Definition 10. $A \in \mathbb{R}^{n\times n}$ is **invertible** if $AB = BA = I_n$ for some $B \in \mathbb{R}^{n\times n}$. Such B is unique: $BA = I$, $AB_1 = I \Rightarrow B = B(AB_1) = (BA)B_1 = B_1$ (Lecture 1, Theorem 3(a)). It is the **inverse** A^{-1} of A. For example, if all $d_i \neq 0$, then $\mathrm{diag}(d_1, \ldots, d_n)^{-1} = \mathrm{diag}(d_1^{-1}, \ldots, d_n^{-1})$: both products are $\mathrm{diag}(1, \ldots, 1) = I$ (Lecture 1, Proposition 13(b)).

Theorem 11. Let $A, B \in \mathbb{R}^{n\times n}$ be invertible.

(a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.

(b) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$; more generally $(A_1 \cdots A_k)^{-1} = A_k^{-1} \cdots A_1^{-1}$ for invertible $A_1, \ldots, A_k \in \mathbb{R}^{n\times n}$.

(c) A' is invertible and $(A')^{-1} = (A^{-1})'$.

Proof.

(a) Definition 10:

$$AA^{-1} = A^{-1}A = I \implies A \text{ is the inverse of } A^{-1}$$

(b) Lecture 1, Theorem 3(a):

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I, \qquad (B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$$

$$(A_1 \cdots A_k)^{-1} = \big((A_1 \cdots A_{k-1})A_k\big)^{-1} = A_k^{-1}(A_1 \cdots A_{k-1})^{-1} = \cdots = A_k^{-1} \cdots A_1^{-1}$$

(c) $(AB)' = B'A'$, $I' = I$ (Lecture 1, Theorem 8):

$$A'(A^{-1})' = (A^{-1}A)' = I' = I, \qquad (A^{-1})'A' = (AA^{-1})' = I$$

□

Corollary 12. If $A \in \mathbb{R}^{n\times n}$ is symmetric and invertible, then A^{-1} is symmetric.

Proof.

$A' = A$ (Lecture 1, Definition 7); Theorem 11(c):

$$(A^{-1})' = (A')^{-1} = A^{-1}$$

□

Definition 13. An **elementary matrix** is $E = \omega(I_m)$, obtained from I_m by one elementary row operation ω. For $r \neq s$: $D_r(c)$ ($c \neq 0$) is I_m with entry (r,r) replaced by c (from $R_r \leftarrow cR_r$); $T_{rs}(c)$ is I_m with entry (r,s) replaced by c (from $R_r \leftarrow R_r + cR_s$); P_{rs} is I_m with rows r, s interchanged (from $R_r \leftrightarrow R_s$). For example, for $m = 4$:

$$P_{23} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad T_{31}(c) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ c & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad D_3(c) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Theorem 14. Let ω be an elementary row operation on matrices with m rows and $E = \omega(I_m)$.

(a) $\omega(A) = EA$ for every $A \in \mathbb{R}^{m\times n}$.

(b) E is invertible, and E^{-1} is the elementary matrix of the inverse operation: $D_r(c)^{-1} = D_r(c^{-1})$, $T_{rs}(c)^{-1} = T_{rs}(-c)$, $P_{rs}^{-1} = P_{rs}$.

(c) If row operations $\omega_1, \ldots, \omega_k$ (in this order) take A to B, then $B = PA$ with $P = E_k \cdots E_2 E_1$, $E_j = \omega_j(I_m)$, and P is invertible.

Proof.

(a) $\mathrm{row}_i(EA) = \mathrm{row}_i(E)A$ (Lecture 1, Theorem 4(c)); an unchanged row of E is e_i', and $e_i'A = \mathrm{row}_i(A)$ (Lecture 1, Proposition 11(a)); Lecture 1, Theorem 3(a):

$$\begin{aligned} R_r \leftarrow cR_r: &\quad \mathrm{row}_r(EA) = (ce_r')A = c\,\mathrm{row}_r(A) \\ R_r \leftarrow R_r + cR_s: &\quad \mathrm{row}_r(EA) = (e_r' + ce_s')A = \mathrm{row}_r(A) + c\,\mathrm{row}_s(A) \\ R_r \leftrightarrow R_s: &\quad \mathrm{row}_r(EA) = e_s'A = \mathrm{row}_s(A), \quad \mathrm{row}_s(EA) = e_r'A = \mathrm{row}_r(A) \end{aligned}$$

(b) $\tilde{\omega}$: the inverse operation (Theorem 3(a)), $\tilde{E} = \tilde{\omega}(I)$; by (a):

$$E\tilde{E} = \omega(\tilde{\omega}(I)) = I, \qquad \tilde{E}E = \tilde{\omega}(\omega(I)) = I$$

(c) By (a) and Lecture 1, Theorem 3(a); Theorem 11(b) with (b):

$$B = E_k(\cdots(E_2(E_1A))) = (E_k \cdots E_1)A = PA, \qquad P \text{ invertible}$$

□

4. Deciding invertibility and computing inverses

Theorem 15. Let $A \in \mathbb{R}^{n\times n}$ and R an RREF of A. The following are equivalent: (i) A is invertible; (ii) $R = I_n$; (iii) $Ax = 0$ has only the trivial solution; (iv) $Ax = b$ has a solution for every $b \in \mathbb{R}^n$; (v) A is a product of elementary matrices. Moreover, if $BA = I$ or $AB = I$ for some $B \in \mathbb{R}^{n\times n}$, then A is invertible and $B = A^{-1}$.

Proof. $R = PA$, $P = E_k \cdots E_1$ invertible (Theorem 14(c)).

- $R = I$: $A = P^{-1}(PA) = P^{-1}R = P^{-1}$ (Lecture 1, Theorem 3(a)) is $E_1^{-1} \cdots E_k^{-1}$, a product of elementary matrices, hence invertible (Theorems 14(b), 11(b)); $Ax = 0$ has only the trivial solution (Theorem 9(b)); and $x = A^{-1}b$ solves $Ax = b$.
- $R \neq I$: then R has a zero row, hence a zero last row (Definition 4(c)); otherwise $r = n$, $1 \leqslant k_1 < \cdots < k_n \leqslant n$ forces $k_i = i$, $\mathrm{col}_i(R) = e_i$ for every i (Definition 4), and $R = I$.
- $R \neq I$: $Ax = 0$ for some $x \neq 0$ (Theorem 9(b)), so A has no inverse (else $x = A^{-1}Ax = 0$ ✘), and A is not a product of elementary matrices (such a product is invertible: Theorems 14(b), 11(b)). For $b = P^{-1}e_n$: $Ax = b \iff Rx = PAx = Pb = e_n$, whose last equation reads $0 = 1$; so (iv) fails.
- So each of (i)–(v) holds exactly when $R = I$.

Last sentence; (iii) ⇒ (i):

$BA = I:\ Ax = 0 \implies x = BAx = 0 \implies A$ invertible, $B = BAA^{-1} = A^{-1}$

$AB = I:\ B$ invertible, $B^{-1} = A$ (case just shown) $\implies A^{-1} = (B^{-1})^{-1} = B$ (Theorem 11(a))

□

Corollary 16 (Computing A^{-1})**.** Let $A \in \mathbb{R}^{n\times n}$; row-reduce $[A \mid I_n]$ by Gauss–Jordan elimination.

(a) If the left half becomes I_n, the right half is A^{-1}.

(b) If a zero row appears in the left half, A is not invertible.

Proof.

(a) E: the product of the elementary matrices used (Theorem 14(c)); $\mathrm{col}_j(EM) = E\,\mathrm{col}_j(M)$ (Lecture 1, Theorem 4(a)):

$[I \mid B] = E[A \mid I] = [EA \mid E] \implies EA = I \implies B = E = A^{-1}$ (Theorem 15, last sentence)

(b) The left half is EA, E invertible (Theorem 14(c)); $\mathrm{row}_i(EAX) = \mathrm{row}_i(EA)X$ (Lecture 1, Theorem 4(c)):

$\mathrm{row}_i(EA) = 0 \implies \mathrm{row}_i(EAX) = 0 \implies EAX \neq I_n$ for all $X \in \mathbb{R}^{n\times n}$

$\implies EA$ not invertible

A invertible $\implies EA$ invertible ✘ (Theorem 11(b))

□

Proposition 17. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2\times 2}$. Then A is invertible $\iff ad - bc \neq 0$, and then

$$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Proof. Row-reduce $[A \mid I_2]$ and read the result by Corollary 16; $\Delta = ad - bc$.

$a \neq 0$ ($d - \frac{cb}{a} = \frac{\Delta}{a}$; $\frac{1}{a} + \frac{bc}{a\Delta} = \frac{\Delta+bc}{a\Delta} = \frac{d}{\Delta}$):

$$\left(\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - \frac{c}{a}R_1} \left(\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & \frac{\Delta}{a} & -\frac{c}{a} & 1 \end{array}\right) \qquad \Delta = 0 \implies \text{zero row}$$

$$\Delta \neq 0: \xrightarrow[R_2 \leftarrow \frac{a}{\Delta}R_2]{R_1 \leftarrow \frac{1}{a}R_1} \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & 1 & -\frac{c}{\Delta} & \frac{a}{\Delta} \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - \frac{b}{a}R_2} \left(\begin{array}{cc|cc} 1 & 0 & \frac{d}{\Delta} & -\frac{b}{\Delta} \\ 0 & 1 & -\frac{c}{\Delta} & \frac{a}{\Delta} \end{array}\right)$$

$\Delta = 0$: not invertible (Corollary 16(b)); $\Delta \neq 0$: the right half is the stated A^{-1}.

$a = 0$, so $\Delta = -bc$; $c = 0$: $Ae_1 = \mathrm{col}_1(A) = 0$ (Lecture 1, Proposition 11(a)), not invertible (Theorem 15, (i)⇔(iii)); $c \neq 0$:

$$\left(\begin{array}{cc|cc} 0 & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|cc} c & d & 0 & 1 \\ 0 & b & 1 & 0 \end{array}\right) \quad b = 0 \implies \text{zero row, not invertible (Corollary 16(b))}$$

$$b \neq 0: \xrightarrow[R_2 \leftarrow \frac{1}{b}R_2]{R_1 \leftarrow \frac{1}{c}R_1} \left(\begin{array}{cc|cc} 1 & \frac{d}{c} & 0 & \frac{1}{c} \\ 0 & 1 & \frac{1}{b} & 0 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - \frac{d}{c}R_2} \left(\begin{array}{cc|cc} 1 & 0 & -\frac{d}{bc} & \frac{1}{c} \\ 0 & 1 & \frac{1}{b} & 0 \end{array}\right)$$

The right half is $\frac{1}{\Delta}\begin{pmatrix} d & -b \\ -c & 0 \end{pmatrix}$, the stated A^{-1} for $a = 0$.

So A is invertible exactly when $\Delta \neq 0$ (for $a = 0$: $b \neq 0$, $c \neq 0$), with the stated inverse (Corollary 16(a)). □

Check by multiplication: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

Example 18. $A = \left(\begin{smallmatrix} 3 & 5 \\ 1 & 2 \end{smallmatrix}\right)$: $ad - bc = 6 - 5 = 1$, so $A^{-1} = \left(\begin{smallmatrix} 2 & -5 \\ -1 & 3 \end{smallmatrix}\right)$ (Proposition 17). $B = \left(\begin{smallmatrix} 2 & 6 \\ 1 & 3 \end{smallmatrix}\right)$: $ad - bc = 6 - 6 = 0$, not invertible; indeed $x = (3, -1)' \neq 0$ has $Bx = (6-6,\ 3-3)' = 0$ (Theorem 15, (i)$\Leftrightarrow$(iii)).

Example 19. Find the inverse of $A = \left(\begin{smallmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -2 & 1 \\ 2 & -1 & 2 & -1 \\ 1 & -1 & 1 & 0 \end{smallmatrix}\right)$, and write A and A^{-1} as products of elementary matrices.

Solution.

Forward elimination of $[A \mid I_4]$:

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ -1 & 1 & -2 & 1 & 0 & 1 & 0 & 0 \\ 2 & -1 & 2 & -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

Entry $(2, 2)$ is 0 after the first arrow: Step 2 interchanges rows 2, 3; Step 3 multiplies row 3 by -1.

Backward elimination, clearing columns 4, 3, 2; Corollary 16(a):

$$\xrightarrow[R_2 \leftarrow R_2 - R_4]{R_1 \leftarrow R_1 + R_4} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 + R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right), \qquad A^{-1} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

Check: $\text{row}_3(A)A^{-1} = (2, -1, 2, -1)A^{-1} = (0+1-2+1,\ 2+0-2+0,\ 2-1+0+0,\ 0+1+0-1) = (0, 0, 1, 0)$; similarly for the other rows.

The nine operations; Theorems 14(c), 15 (last sentence), 11(a),(b), 14(b):

$$E_1 = T_{21}(1),\ E_2 = T_{31}(-2),\ E_3 = T_{41}(-1),\ E_4 = P_{23},\ E_5 = D_3(-1)$$
$$E_6 = T_{14}(1),\ E_7 = T_{24}(-1),\ E_8 = T_{13}(-1),\ E_9 = T_{12}(1)$$
$$E_9E_8 \cdots E_1A = I_4 \implies A^{-1} = E_9E_8 \cdots E_1$$
$$A = (A^{-1})^{-1} = E_1^{-1}E_2^{-1} \cdots E_9^{-1} = T_{21}(-1)\,T_{31}(2)\,T_{41}(1)\,P_{23}\,D_3(-1)\,T_{14}(-1)\,T_{24}(1)\,T_{13}(1)\,T_{12}(-1)$$

Example 20. The matrix A of Example 8 is not invertible: an RREF of A, the left part of the last matrix there (Definition 2), is $\left(\begin{smallmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right) \neq I_4$. Accordingly $Ax = 0$ has the nontrivial solution $x = (-2, 1, 1, 0)'$, and $Ax = y$ has no solution for $y = (1, 0, 0, 0)'$ (Theorem 15, (iii) and (iv)).

5. Solving $Ax = b$

Theorem 21. Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and let row operations take $[A \mid b]$ to $[H \mid d]$ in echelon form ($H \in \mathbb{R}^{m\times n}$). Then $Ax = b$ is consistent $\iff$ $[H \mid d]$ has no row $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$ $\iff$ the last column of $[H \mid d]$ is not a pivot column. If row operations take $[A \mid b]$ to any matrix with a row $(0 \ \cdots \ 0 \mid c)$, $c \neq 0$, then $Ax = b$ has no solution. If $[H \mid d]$ has no such row, let x_j be a **pivot unknown** if column j of H is a pivot column, a **free unknown** otherwise (as in Proposition 7, now for H): each choice of values of the free unknowns gives, by **back substitution** (solving the equations of the nonzero rows of H, from the last one up, for their pivot unknowns), exactly one solution, and every solution arises so.

Proof.

- $Ax = b$ and $Hx = d$ have the same solutions (Theorem 3(b)). In an echelon form, a row $(0 \ \cdots \ 0 \mid c)$, $c \neq 0$, is exactly a row whose pivot lies in the last column.
- If $[B \mid z]$, reached from $[A \mid b]$, has such a row, the equation $0 = c$ of $Bx = z$ has no solution; nor has $Ax = b$ (Theorem 3(b)). So one may stop at any stage of the elimination.
- No such row in $[H \mid d]$: each zero row of H gives $0 = 0$. Let rows $1, \dots, r$ of $H = (h_{ij})$ be nonzero, with pivots in columns $k_1 < \cdots < k_r$.
- Fix the free unknowns. Row i of H is 0 before column k_i and 1 there, so for $i = r, r-1, \dots, 1$ equation i reads $x_{k_i} + \sum_{j>k_i} h_{ij}x_j = d_i$; each x_j, $j > k_i$, is free or already found. So it determines x_{k_i}, and x solves $Hx = d$, hence $Ax = b$.
- The equations force the pivot unknowns: every solution arises from its free unknowns, exactly once. □

Theorem 22. Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$.

(a) If $Ax_p = b$ (a **particular solution**), the solutions of $Ax = b$ are exactly the $x_p + z$ with $Az = 0$.

(b) $Ax = b$ has no solution, exactly one, or infinitely many. If it is consistent: exactly one $\iff$ no free unknowns (every column of H in Theorem 21 is a pivot column).

(c) If $m = n$: $Ax = b$ has exactly one solution $\iff$ A is invertible; this solution is $x = A^{-1}b$.

Proof.

(a) Lecture 1, Theorem 3(a):

$$\begin{aligned} Az = 0 &\implies A(x_p + z) = Ax_p + Az = b \\ Ax = b &\implies z = x - x_p : \ Az = b - b = 0, \ x = x_p + z \end{aligned}$$

(b)

- Let $Ax = b$ be consistent: forward elimination (Definition 5, Steps 1–4) takes $[A \mid b]$ to an echelon form $[H \mid d]$, with no row $(0 \ \cdots \ 0 \mid c)$, $c \neq 0$ (Theorem 21).
- By Theorem 21, each choice of the free unknowns gives exactly one solution, and every solution arises so; different choices give different solutions (they differ in a free unknown, an entry of x).
- No free unknowns: exactly one solution; a free unknown: each of its values gives another solution, infinitely many.

(c) Lecture 1, Theorem 3(a):

$$\begin{aligned} A \text{ invertible} &\implies A(A^{-1}b) = b; \quad Ax = b \implies x = A^{-1}(Ax) = A^{-1}b \\ \text{one solution } x_p &\implies (Az = 0 \implies z = 0) \quad (\text{(a): else } x_p + z \text{ is a second solution}) \\ &\implies A \text{ invertible} \quad (\text{Theorem 15, (iii)}\Rightarrow\text{(i)}) \end{aligned}$$

□

Example 23. Solve $x_1 - x_2 + x_3 - x_4 = 1$, $-x_1 + x_2 - 2x_3 + x_4 = 0$, $2x_1 - x_2 + 2x_3 - x_4 = 3$, $x_1 - x_2 + x_3 = 3$.

Solution.

A of Example 19, $b = (1, 0, 3, 3)'$; forward elimination of $[A \mid b]$ by the operations there:

$$\left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ -1 & 1 & -2 & 1 & 0 \\ 2 & -1 & 2 & -1 & 3 \\ 1 & -1 & 1 & 0 & 3 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

Echelon form, no row $(0\ 0\ 0\ 0 \mid c)$, $c \neq 0$: consistent (Theorem 21); no free unknowns: one solution (Theorem 22(b)):

$$x_4 = 2, \quad x_3 = -1, \quad x_2 = 1 - x_4 = -1, \quad x_1 = 1 + x_2 - x_3 + x_4 = 1 - 1 + 1 + 2 = 3$$

$x = (3, -1, -1, 2)'$; check with A^{-1} of Example 19 (Theorem 22(c)):

$$A^{-1}b = (0 + 0 + 3 + 0,\ -1 + 0 + 3 - 3,\ -1 + 0 + 0 + 0,\ -1 + 0 + 0 + 3)' = (3, -1, -1, 2)'$$

Example 24. Solve $x_1 + x_2 + x_3 - 2x_4 = 2$, $x_1 + x_2 + 2x_3 - 3x_4 = 3$, $2x_1 + x_2 + 5x_3 - 4x_4 = 7$.

Solution.

A of Example 6, $b = (2, 3, 7)'$; forward elimination of $[A \mid b]$ by the operations there:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 1 & 1 & 2 & -3 & 3 \\ 2 & 1 & 5 & -4 & 7 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & -1 & 3 & 0 & 3 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 0 & -1 & 3 & 0 & 3 \\ 0 & 0 & 1 & -1 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 0 & 1 & -3 & 0 & -3 \\ 0 & 0 & 1 & -1 & 1 \end{array}\right)$$

Echelon form, no row $(0\ 0\ 0\ 0 \mid c)$, $c \neq 0$: consistent (Theorem 21); pivot columns $1, 2, 3$, $x_4 = t$ free: infinitely many solutions (Theorem 22(b)):

$$x_3 = 1 + t, \quad x_2 = -3 + 3x_3 = 3t, \quad x_1 = 2 - x_2 - x_3 + 2x_4 = 2 - 3t - (1 + t) + 2t = 1 - 2t$$

$$x = (1, 0, 1, 0)' + t(-2, 3, 1, 1)' \quad (t \in \mathbb{R})$$

This is the particular solution $(1, 0, 1, 0)'$ plus the solutions $t(-2, 3, 1, 1)'$ of $Ax = 0$ found in Example 6, as Theorem 22(a) says.

Practice. The Exercises and Problems of the textbook, Chapter 2: row reduction and reduced row echelon forms, reading off the solutions, inverses by Gauss–Jordan elimination, elementary matrices, and solving $Ax = b$.